

Experiment – 03

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Subject: Advanced Database
Management System

Question 1:

You are given with the Employee schema; the task is to find the highest score for each respective department. For eg: department D1 have the highest score as 5.28. Hence in the output, final updated score for D1 should be 5.28. Same with the other departments i.e., D2, D3 and D4.

EID	DEPT	SCORES
1	D1	1
2	D1	5.28
3	D1	4
4	D2	8
5	D1	2.5
6	D2	7
7	D3	9
8	D4	10.2

Output:

EID	DEPT	SCORES
1	D1	5.28
2	D1	5.28
3	D1	5.28
4	D2	8
5	D1	5.28
6	D2	8
7	D3	9

Solution:

```
UPDATE EMPLOYEE AS E1
SET SCORES = (
    SELECT MAX(SCORES)
    FROM EMPLOYEE
    WHERE DEPT = E1.DEPT
);
```

Question 2:

A company maintains records of products sold by different salespersons. The management wants to prepare a sales performance report by identifying the salesperson(s) who generated the highest overall sales revenue. If multiple salespersons have the same highest total sales amount, all of them must be included in the final report. Return the seller ID(s) satisfying this condition.

SalesPerson				Orders			Company		
seller_id	name	city	commission	order_id	seller_id	amount	com_id	name	city
1	John	New York	15	1	1	1200	1	RED	Boston
2	Amy	Los Angeles	13	2	1	800	2	ORANGE	New York
3	Mark	Chicago	12	3	2	2500	3	YELLOW	Boston
4	Pam	Boston	15	4	3	1500	4	GREEN	Austin
				5	2	700			
				6	3	2000			

Output:

seller_id
3

Solution:

```

SELECT SELLER_ID
FROM ORDERS
GROUP BY SELLER_ID
HAVING SUM(AMOUNT) = (
    SELECT MAX(MAX_AMT)
    FROM (
        SELECT
            SELLER_ID,
            SUM(AMOUNT) AS MAX_AMT
        FROM ORDERS
        GROUP BY SELLER_ID
    ) AS T
);

```

Questio 3:

A company maintains employee salary records for different departments. The management wants to prepare a report containing employees whose salaries fall among the top three highest distinct salary values within their respective departments. If multiple employees receive the same qualifying salary, all such employees must be included in the final report. Display the department name, employee name, and salary for every qualifying employee.

ID	Name
1	IT
2	Sales

Output:

Department	Employee	Salary
IT	Max	90000
IT	Joe	85000
IT	Randy	85000
IT	Will	70000
Sales	Henry	80000
Sales	Sam	60000

ID	Name	Salary	DepartmentId
1	Joe	85000	1
2	Henry	80000	2
3	Sam	60000	2
4	Max	90000	1
5	Janet	69000	1
6	Randy	85000	1
7	Will	70000	1

Solution:

```

SELECT
    D.NAME AS DEPARTMENT,
    E.NAME AS EMPLOYEE,

```

```
E.SALARY AS SALARY
FROM EMPLOYEE AS E
JOIN DEPARTMENT AS D
  ON E.DEPARTMENTID = D.ID
WHERE (
  SELECT COUNT(DISTINCT E2.SALARY)
  FROM EMPLOYEE AS E2
  WHERE E2.DEPARTMENTID = E.DEPARTMENTID
  AND E2.SALARY > E.SALARY
) < 3
ORDER BY D.NAME;
```